

Knowledge Representation

by Adia Khalid
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Lecture notes by Liliana Sanfilippo

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Part I

Foundation of Knowledge Representation

1

Introduction to KR: Definition, Scope, Motivation (Use Cases), and History

1.1 Introduction

What is Knowledge Representation?

Knowledge Representation

Definition 1.1 KR is a central problem in Artificial Intelligence that focuses on developing sufficiently precise notations for representing knowledge

The core problem with KR is that the real world with its complexity, ambiguity and dynamism can never be fully represented in a computer-based system. Nevertheless, intelligent systems must be able to function effectively using partial, incomplete and sometimes contradictory information.

Why is Knowledge Representation needed?

The problem of data meaninglessness

To understand the need for KR, let's consider a practical scenario:

Example 1.1 A student database.

A hypothetical database contains the following entries:

Name	Major	CGPA	Research	...
Alice Smith	CS	3.8	Genomics	...
Bob John	Biology	3.8	Oncology	...
Carol Li	Zoology	3.7	Animal colonisation	...

Without knowledge representation, this data is disjointed. The entries exist in isolation across thousands of files. A database query system can only perform exact keyword matches:

- Query 1: "Show me all AI researchers" → No results (not explicitly mentioned in the data)
- Query 2: "Which students could work on this patent?" → No answer possible
- Query 3: "What are the emerging trends?" → No analysis possible

Data is meaningless without context and relationships. It does not allow for intelligent reasoning or deeper analysis.

What knowledge representation makes possible

With knowledge representation, we can structure the data and transfer isolated records into interconnected, queryable knowledge using:

- **ontologies**: Explicitly specify which concepts exist (student, researcher, field of research) and how they are categorised.
- Knowledge graphs: Link entities together: Alice → researcher → AI

This allows for semantic searches, automatic relationship discovery and reasoning across all connection.

Example 1.2 • Semantic search

- “Find students researching machine learning?”
- “Show me all AI researchers”: The system can use transitive relationships to find relevant students.
- “Which students are a good fit for this patent?": The system can match skills and areas of interest.
- Automatic relationship discovery
 - “What are emerging research trends?”
 - Recognising patterns across all data
 - Identifying emerging trends through interconnected information

Data	Information	Knowledge
Raw, unprocessed facts	Structured and contextualised data	Information combined with experience and reasoning rules
e.g. CGPA: 3.8	e.g. Alice has a CGPA of 3.8 in Computer Science	e.g. Alice could work on this AI patent because she has a strong foundation in computer science and is interested in genomics

Table 1.1: Data vs. Information vs. Knowledge

KR enables machines to make informed decisions with fewer rules.

KR Perspective

Representation Schemes

Challenges in Knowledge Representation

Traditional ML systems follow a black-box approach which means the decision-making process is not transparent and the reasoning behind predictions is not accessible.

- Generalisation: Requires continuous retraining with new data
- Explainability: Difficult or impossible to explain decisions
- Data flexibility: Highly dependent on large amounts of training data

Knowledge-Based AI utilises structured, **explicit knowledge**.

explicit knowledge

Definition 1.2 Knowledge that is formally encoded in symbols, rules or graphs and can be processed by machines

- Generalization: explicit knowledge transfers to new problems without retraining from scratch[2]
- Explainability: encoded knowledge enables transparent reasoning and human-understandable decisions[1]
- Data flexibility: Modifiable

Representation Schemes

representation scheme

Definition 1.3 A representation scheme is a formal notation used to define a knowledge base (Mylopoulos 1980). A knowledge base contains facts and rules and serves as a model of the real world. A representation scheme is therefore the ‘language’ in which we express knowledge, much like programming languages define the syntax and semantics in which we write instructions.

Example 1.3 Pac-Man ghost AI has a knowledge-based approach: We code ghost behaviour as explicit rules.

```
IF player_nearby THEN chase
IF frightened THEN flee
ELSE patrol
```

Decisions are transparent, explainable, and modifiable without recomputing. (Øgland 2016)

Challenges in Knowledge Representation

1.2 Taxonomy of Representation Schemes

[3]

1.3 Distinguishing Features of KR Systems

Defis: Frame, Inference,
kinds of reasoning inference types (23+) David, Shrobe and Szolovist
Wie man mit Incompleteness umgeht

1.4 Applications of Knowledge Representation

Enterprise Knowledge Management

1.5 Future Directions

Multi-hop reasoning Unified framework
knowledge Aggregation
def attention weight

1.6 Course structure

foundation of KR
Knowledge graphs
KR in mL
grading:
50% of the exercise points (full portfolio) (Abgabe in 3er Gruppen(?)) present one exercise in tut present one paper in seminar 40–45 min (3er gruppen) final report 5–10 pages (50% marks on final report)
Es wird ein Moodle geben

1.7 Formalities

2

Logical Foundations: First-Order Logic (FOL)

3

Logic Programming and Automated Reasoning

4

Structured Knowledge Representation

Part II

Knowledge Graphs

5

Start into semantic networks and knowledge graphs reasoning

6

The Resource Description Framework (RDF)

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Knowledge Graph Reasoning



Knowledge Acquisition from Data

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Representation Learning

Part III

Knowledge Representation in Machine Learning

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Knowledge Graph Embeddings

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Graph Neural Networks for Knowledge Representation

Part IV

Applications

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Knowledge Tracing

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Knowledge in Large Language Models

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KG- Aware Applications

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